

Anesthesia for Trauma and Burns MCQ – Chapter 35

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 35)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Why is management of trauma patients uniquely challenging?

- A. Inadequate medical history, not fasting, intoxication, multiple injuries, shock and difficult airway
- B. Only due to blood loss
- C. Only due to intoxication
- D. Only due to multiple injuries

Ans: A

Q2. Why is adequate medical history often unavailable in trauma patients?

- A. Language barrier only
- B. Patient refuses to share
- C. Hospital policy
- D. Attendant may not be present and patient may not be fit enough to give history

Ans: D

Q3. What is the minimum systolic blood pressure recommended before induction in trauma patients?

- A. 80-90 mm Hg
- B. 120 mm Hg
- C. 60-70 mm Hg
- D. 100-110 mm Hg

Ans: A

Q4. When may damage control surgery warrant induction at very low blood pressure?

- A. When surgery is the only method to stop bleeding
- B. When patient requests it
- C. For elective procedures
- D. For cosmetic surgery

Ans: A

Q5. What major injuries must be ruled out in trauma patients?

- A. Only abdominal injuries
- B. Chest injury (pneumo/hemothorax), head injury, spine injuries, abdominal and pelvic injuries
- C. Only spine injuries
- D. Only chest injuries

Ans: B

Q6. Why is head injury significant for choice of anesthesia technique?

- A. It makes GA easier
- B. Central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT)
- C. It has no effect
- D. It only affects monitoring

Ans: B

Q7. What devastating complications can improper movements of spine injury patients cause? A. Only numbness B. Only pain C. Quadriplegia, paraplegia or even death D. Only temporary weakness	Ans: C
Q8. If fasting status cannot be obtained in trauma, what premedication should be given? A. Only omeprazole B. Only antacids C. Metoclopramide and ranitidine/sodium citrate D. Only ondansetron	Ans: C
Q9. Why is continuous blood pressure by invasive monitoring recommended in trauma? A. Although difficult to cannulate artery in low output state, every effort should be made as trauma patients are prone to hypotension B. It is cheaper C. For research purposes D. It is more convenient	Ans: A
Q10. Why is temperature monitoring essential in trauma patients? A. For billing purposes B. To calculate drug doses C. Because trauma patients are very prone to hypothermia D. For documentation	Ans: C
Q11. How should a trauma patient always be considered regarding stomach contents? A. As safe for LMA B. As having a full stomach until proved otherwise, managed like rapid sequence induction C. As partially fasted D. As having an empty stomach	Ans: B
Q12. What is the choice of anesthesia for majority of trauma surgeries? A. Monitored anesthesia care B. Local anesthesia C. General anesthesia (except for limb saving surgeries which may be done under regional) D. Regional anesthesia	Ans: C
Q13. Why is selection of induction agent difficult in trauma patients? A. Only availability issues B. Only cost issues C. IV induction agents may produce profound hypotension or even cardiac arrest in shock patients D. All agents are equally safe	Ans: C

Q14. How is the dose requirement of IV induction agents in hypotensive patients? A. Doubled B. Increased C. Same as normal D. Drastically reduced	Ans: D
Q15. Which agents are often selected for induction in trauma patients and why? A. Etomidate (cardiovascular stability) and ketamine (sympathetic system stimulating property) B. Propofol for its rapid onset C. Midazolam for its amnesia D. Thiopentone for its potency	Ans: A
Q16. Why can etomidate still cause hypotension in catecholamine-depleted shock patients? A. It causes direct cardiac depression B. It causes adrenocortical suppression C. It causes histamine release D. It is a vasodilator	Ans: B
Q17. Why can ketamine cause hypotension in catecholamine-depleted patients? A. It causes histamine release B. It causes direct myocardial depression in the absence of catecholamine C. It blocks sympathetic ganglia D. It causes vasodilation	Ans: B
Q18. What should be assumed about ALL trauma patients regarding cervical spine? A. Cervical spine is safe B. Only elderly patients C. All should be assumed to have cervical spine injury until proved otherwise D. Only high-velocity trauma patients	Ans: C
Q19. How should trauma patients with uncleared cervical spine be intubated? A. Without any precautions B. Only with tracheostomy C. Blind nasal intubation D. With in-line manual stabilization	Ans: D
Q20. When should nasal route NOT be acquired for intubation in trauma? A. In all cases B. Until basal skull fracture is ruled out C. In mandibular fractures D. In neck trauma only	Ans: B

Q21. What equipment must be ready for difficult airway in trauma including face/neck trauma? A. Only LMA B. Only bougie C. All equipment for difficult airway including emergency cricothyrotomy D. Only extra ETT	Ans: C
Q22. Through which route can patients with maxillary fractures most often be easily intubated? A. Oral route B. Tracheostomy only C. LMA only D. Nasal route	Ans: A
Q23. Why are patients with mandibular fracture often intubated nasally? A. For better surgical access only B. Oral intubation is preferred C. Trismus prevents oral intubation and maxillofacial surgeons do dental wiring to close mouth after surgery D. Nasal route is easier	Ans: C
Q24. What may patients with both mandibular and maxillary fracture require? A. Nasal intubation B. Tracheostomy C. Oral intubation D. LMA	Ans: B
Q25. What is the best solution for pharyngeal and laryngeal trauma where intubation can worsen the injury? A. A low tracheostomy B. LMA C. Oral intubation D. Nasal intubation	Ans: A
Q26. Is application of cricoid pressure contraindicated in any trauma situation? A. Only in cervical spine injury B. Only in facial fractures C. Yes, it is contraindicated in laryngeal trauma D. Never contraindicated	Ans: C
Q27. Why must a chest tube be inserted before intubation if there is pneumothorax? A. Because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax B. To prevent bleeding C. To improve oxygenation D. To allow nasal intubation	Ans: A

Q28. Which inhalational agent is most preferred for maintenance in trauma patients? A. Halothane B. Sevoflurane C. Isoflurane D. Desflurane	Ans: D
Q29. Why may patients in shock exhibit extreme sensitivity to inhalational agents? A. Like IV agents, patients in shock exhibit extreme sensitivity to inhalational agents B. Due to hepatic failure C. Due to hypothermia only D. Due to renal failure	Ans: A
Q30. Despite causing hyperkalemia in crush injury, why does succinylcholine remain the muscle relaxant of choice in trauma? A. Because it remains the choice for intubation in difficult airway and high risk of aspiration - consequences of failed intubation and aspiration can be devastating B. It causes no side effects C. It is the cheapest D. It has the longest duration	Ans: A
Q31. Which non-depolarizing muscle relaxants are preferred in trauma? A. Rocuronium only B. Vecuronium and pancuronium C. Atracurium and Cis-atracurium (due to their independency on hepatic and renal functions) D. Mivacurium only	Ans: C
Q32. What type of IV access should be secured in trauma patients? A. Only central line B. Only intraosseous C. Small peripheral cannula D. Large bore cannulas; if peripheral access not possible then central access (most often subclavian) or intraosseous access	Ans: D
Q33. How should hypothermia be prevented in trauma patients? A. Only warm fluids B. Only warming blankets C. OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion D. No measures needed	Ans: C
Q34. Why is extubation tricky in trauma patients? A. Only due to pain B. Deep extubation can cause aspiration while light extubation can increase intraocular/intracranial pressure and restart hemorrhage C. Only due to swelling D. Only due to bleeding	Ans: B

Q35. Is it uncommon to see trauma patients getting electively ventilated postoperatively? A. Only in burns B. Only in head injury C. No, it is not uncommon D. Yes, very uncommon	Ans: C
Q36. What is the most common immediate cause of death if burns occur in closed space? A. Hypovolemia B. Hypothermia C. Carbon monoxide poisoning D. Sepsis	Ans: C
Q37. What is the immediate treatment for carbon monoxide poisoning in burns? A. IV fluids B. Oxygen or hyperbaric oxygen (in severe cases) C. Intubation D. Steroids	Ans: B
Q38. What formula is used for fluid replacement in burn patients? A. Holliday formula B. 4-2-1 rule C. Parkland formula (4 mL/kg/% burn) D. Holiday-Segar formula	Ans: C
Q39. How should the Parkland formula fluid be distributed? A. All in first 4 hours B. Equal over 24 hours C. All in first 8 hours D. 50% in first 8 hours and next 50% in next 16 hours	Ans: D
Q40. What is the current recommendation for fluid management in burns? A. Restrict fluids B. Give only colloids C. Start fluid by Parkland formula but immediately start tapering once urine output becomes >0.5-1 mL/kg/hr D. Give only crystalloids	Ans: C
Q41. What is the fluid of choice in burn resuscitation? A. Dextrose B. Normal saline C. Ringer lactate D. Colloids	Ans: C

Q42. Can colloids be given in acute phase of burns? A. Yes, if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours) B. Absolutely contraindicated C. Only albumin D. Only after 72 hours	Ans: A
Q43. What cardiac complications can electrical burns present with? A. Only hypertension B. Cardiac arrhythmias or even cardiac arrest C. Only bradycardia D. Only tachycardia	Ans: B
Q44. Why is intraoperative ECG monitoring mandatory in electrical burns? A. It is not mandatory B. For billing C. For research D. Because there are high chances of arrhythmias	Ans: D
Q45. What is the usual surgery done in burn patients? A. Amputation B. Release of post-burn contractures and split skin grafts C. Debridement only D. Fasciotomy only	Ans: B
Q46. Why is succinylcholine contraindicated in burn patients and for how long? A. For one year due to presence of extrajunctional receptors causing hyperkalemia B. For 6 months due to bradycardia C. For 2 weeks due to pain D. It is not contraindicated	Ans: A
Q47. Why may the dose of IV induction agent be slightly higher in burn patients? A. Due to renal failure B. Due to hepatic enzyme induction C. Due to large volume of distribution (hyperdynamic circulation) and increased metabolism D. Due to tolerance	Ans: C
Q48. Why may airway be difficult in burn patients? A. Due to trismus B. Due to laryngeal injury only C. Due to edema only D. Due to contracture of face and neck	Ans: D

Q49. If fiberoptic intubation is not possible in burn contracture patients, what alternative is used?

- A. LMA only
- B. Tracheostomy always
- C. Release contracture under IV ketamine and then intubation is done
- D. Blind nasal intubation

Ans: C

Q50. What should be evaluated in patients with electrical burns before giving regional anesthesia?

- A. Neurological deficits (as electrical burns can have neurological injuries)
- B. Blood pressure
- C. Skin condition
- D. Temperature

Ans: A

Answer Key & Explanations

Q1. Answer: A — Inadequate medical history, not fasting, intoxication, multiple injuries, shock and difficult airway

Management of trauma patients poses unique challenges due to inadequate medical history, most patients are not fasting, may be intoxicated, multiple injuries make anesthesia complex, airway may be difficult due to facial/neck injuries, and presence of shock.

Topic: Trauma General

Q2. Answer: D — Attendant may not be present and patient may not be fit enough to give history

Inadequate medical history: Attendant may not be present and patient may not be fit enough to give his/her medical history.

Topic: Trauma General

Q3. Answer: A — 80-90 mm Hg

A systolic blood pressure of at least 80-90 mm Hg is recommended before induction.

Topic: Preoperative

Q4. Answer: A — When surgery is the only method to stop bleeding

Sometimes the situation like damage control surgery (surgery is the only method to stop bleeding) warrants the induction at very low or even nonrecordable blood pressure.

Topic: Preoperative

Q5. Answer: B — Chest injury (pneumo/hemothorax), head injury, spine injuries, abdominal and pelvic injuries

It is very important to rule out major injuries especially chest injury like pneumo/hemothorax, head injury, spine injuries, and abdominal/pelvic injuries.

Topic: Preoperative

Q6. Answer: B — Central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT)

Head injury—presence of head injury can alter the anesthesia technique. For example, central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT).

Topic: Preoperative

Q7. Answer: C — Quadriplegia, paraplegia or even death

Spine injuries: Improper movements can produce devastating complications, including quadriplegia, paraplegia or even death.

Topic: Preoperative

Q8. Answer: C — Metoclopramide and ranitidine/sodium citrate

If fasting status cannot be obtained, pre-medicate the patient with metoclopramide and ranitidine/sodium citrate.

Topic: Preoperative

Q9. Answer: A — Although difficult to cannulate artery in low output state, every effort should be made as trauma patients are prone to hypotension

Although it may be difficult to cannulate artery in low output state but still every effort should be made to measure continuous blood pressure by invasive monitoring.

Topic: Monitoring

Q10. Answer: C — Because trauma patients are very prone to hypothermia

As trauma patients are very prone for hypothermia, temperature monitoring becomes must.

Topic: Monitoring

Q11. Answer: B — As having a full stomach until proved otherwise, managed like rapid sequence induction

A trauma patient should always be considered full stomach until proved otherwise and should be managed like rapid sequence induction.

Topic: Choice of Anesthesia

Q12. Answer: C — General anesthesia (except for limb saving surgeries which may be done under regional)

Except for limb saving surgeries which may be performed in regional anesthesia, majority of the trauma surgeries are done in general anesthesia.

Topic: Choice of Anesthesia

Q13. Answer: C — IV induction agents may produce profound hypotension or even cardiac arrest in shock patients

Selection of the induction agent is difficult in trauma patients. IV induction agents may produce profound hypotension (or even cardiac arrest) in shock patients.

Topic: Induction

Q14. Answer: D — Drastically reduced

Requirement of doses is drastically reduced in hypotensive patients.

Topic: Induction

Q15. Answer: A — Etomidate (cardiovascular stability) and ketamine (sympathetic system stimulating property)

Etomidate (because of its cardiovascular stability) and ketamine (because of its sympathetic system stimulating property) are the agents often selected for induction.

Topic: Induction

Q16. Answer: B — It causes adrenocortical suppression

Etomidate can produce hypotension in shock patients who are already catecholamine depleted; etomidate by causing adrenocortical suppression.

Topic: Induction

Q17. Answer: B — It causes direct myocardial depression in the absence of catecholamine

Ketamine can produce hypotension in shock patients who are already catecholamine depleted; ketamine by causing direct myocardial depression in the absence of catecholamine.

Topic: Induction

Q18. Answer: C — All should be assumed to have cervical spine injury until proved otherwise

All trauma patients should be assumed to be having cervical spine injury until proved otherwise.

Topic: Intubation

Q19. Answer: D — With in-line manual stabilization

If cervical spine injury is not cleared they should be intubated with in-line manual stabilization.

Topic: Intubation

Q20. Answer: B — Until basal skull fracture is ruled out

Nasal route should not be acquired for intubation until basal skull fracture is ruled out.

Topic: Intubation

Q21. Answer: C — All equipment for difficult airway including emergency cricothyrotomy

All equipment for difficult airway (including emergency cricothyrotomy) must be ready hand available because intubating a patient with face and neck trauma can become a big challenge.

Topic: Intubation

Q22. Answer: A — Oral route

Patients with maxillary fractures should not be intubated nasally; however most often can be easily intubated through oral route.

Topic: Intubation

Q23. Answer: C — Trismus prevents oral intubation and maxillofacial surgeons do dental wiring to close mouth after surgery

Patients with mandibular fracture are often intubated nasally due to two reasons: trismus prevents oral intubation, and most often maxillofacial surgeons do the dental wiring to close the mouth after mandibular surgery.

Topic: Intubation

Q24. Answer: B — Tracheostomy

Patients with both mandibular and maxillary fracture may require tracheostomy.

Topic: Intubation

Q25. Answer: A — A low tracheostomy

Intubation in pharyngeal and laryngeal trauma can worsen the injury or even lead to complete loss of airway. A low tracheostomy is the best solution in most of the cases.

Topic: Intubation

Q26. Answer: C — Yes, it is contraindicated in laryngeal trauma

Application of cricoid pressure is also contraindicated in laryngeal trauma.

Topic: Intubation

Q27. Answer: A — Because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax

Chest tube must be inserted before intubation if there is pneumothorax because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax.

Topic: Intubation

Q28. Answer: D — Desflurane

Desflurane is often the most preferred inhalational agent because of its ability to maintain blood pressure by causing sympathetic stimulation and least metabolism.

Topic: Maintenance

Q29. Answer: A — Like IV agents, patients in shock exhibit extreme sensitivity to inhalational agents

Like IV agents patients in shock may exhibit extreme sensitivity to inhalational agents.

Topic: Maintenance

Q30. Answer: A — Because it remains the choice for intubation in difficult airway and high risk of aspiration - consequences of failed intubation and aspiration can be devastating

In spite of succinylcholine causing hyperkalemia in crush injury and increasing intraocular and intracranial pressure, it remains the muscle relaxant of choice for intubation in trauma patients with difficult airway and high risk of aspiration.

Topic: Muscle Relaxants

Q31. Answer: C — Atracurium and Cis-atracurium (due to their independency on hepatic and renal functions)

Atracurium and Cis-atracurium are preferred non depolarizers because of their independency on hepatic and renal functions.

Topic: Muscle Relaxants

Q32. Answer: D — Large bore cannulas; if peripheral access not possible then central access (most often subclavian) or intraosseous access

Secure IV lines with large bore cannulas. If peripheral access is not possible, then obtain central access (most often subclavian) or intraosseous access (if central access is also not possible).

Topic: Intraoperative

Q33. Answer: C — OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion

As trauma patients are very prone for hypothermia OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion.

Topic: Intraoperative

Q34. Answer: B — Deep extubation can cause aspiration while light extubation can increase intraocular/intracranial pressure and restart hemorrhage

Extubation in trauma patients is a tricky exercise. Deep extubation can cause aspiration while extubation under light anesthesia can increase intraocular, intracranial pressure and restart the hemorrhage by increasing blood pressure.

Topic: Postoperative

Q35. Answer: C — No, it is not uncommon

It is not uncommon to see trauma patients getting electively ventilated in postoperative period.

Topic: Postoperative

Q36. Answer: C — Carbon monoxide poisoning

Carbon monoxide poisoning is the most common immediate cause of death if burns occur in closed space.

Topic: Burns

Q37. Answer: B — Oxygen or hyperbaric oxygen (in severe cases)

Immediate treatment with oxygen or hyperbaric oxygen (in severe cases) may be required.

Topic: Burns

Q38. Answer: C — Parkland formula (4 mL/kg/% burn)

Fluid should be given by the standard Parkland formula which recommends fluid replacement at a rate of 4 mL/kg/% burn.

Topic: Burns

Q39. Answer: D — 50% in first 8 hours and next 50% in next 16 hours

50% of this fluid is given in first 8 hours and next 50% in next 16 hours.

Topic: Burns

Q40. Answer: C — Start fluid by Parkland formula but immediately start tapering once urine output becomes >0.5-1 mL/kg/hr

Current recommendation is to start fluid by Parkland formula but immediately start tapering once urine output becomes >0.5-1 mL/kg/hr.

Topic: Burns

Q41. Answer: C — Ringer lactate

Ringer lactate is the fluid of choice.

Topic: Burns

Q42. Answer: A — Yes, if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours)

Colloids if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours).

Topic: Burns

Q43. Answer: B — Cardiac arrhythmias or even cardiac arrest

Patients with electrical burns can present with cardiac arrhythmias or even cardiac arrest.

Topic: Burns

Q44. Answer: D — Because there are high chances of arrhythmias

Intraoperative monitoring of ECG is mandatory in electric burns as there are high chances of arrhythmias.

Topic: Burns

Q45. Answer: B — Release of post-burn contractures and split skin grafts

Usual surgeries done in burn patients are release of post-burn contractures and split skin grafts.

Topic: Burns

Q46. Answer: A — For one year due to presence of extrajunctional receptors causing hyperkalemia

Presence of extrajunctional receptors contraindicates the use of succinylcholine for one year and decreases the sensitivity (increase dose requirement) of nondepolarizers.

Topic: Burns

Q47. Answer: C — Due to large volume of distribution (hyperdynamic circulation) and increased metabolism

Dose of intravenous induction agent may be slightly higher because of the large volume of distribution (hyperdynamic circulation) and increased metabolism.

Topic: Burns

Q48. Answer: D — Due to contracture of face and neck

Airway may be difficult due to contracture of face and neck. Awake intubation under fiberoptic bronchoscopy may be required.

Topic: Burns

Q49. Answer: C — Release contracture under IV ketamine and then intubation is done

If that is not possible then contracture is released under intravenous ketamine and then intubation is done.

Topic: Burns

Q50. Answer: A — Neurological deficits (as electrical burns can have neurological injuries)

Patients with electrical burn can have neurological injuries. They should be evaluated for neurological deficits before giving regional anesthesia.

Topic: Burns